Predicting Student Academic Performance Based on Lifestyle and Behavioural Factors

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Problem Statement

Academic success is driven by a complex interplay of intelligence, lifestyle choices, and mental health. For educational institutions, identifying these drivers is critical for improving outcomes and reducing failure rates.

This project investigates whether students' academic performance can be reliably predicted using behavioural and lifestyle data. To address this, machine learning techniques, including regression and classification, are applied to identify key factors influencing performance and support targeted interventions for at-risk students.

Data Summary

The dataset contains approximately 80000 student records and includes multiple features representing academic, behavioural, and lifestyle factors such as study hours, sleep duration, stress levels, and social media usage. The target variable is exam_score, a continuous variable representing final exam performance.

Before analysis, data preprocessing was performed. Missing values were handled using mean imputation, categorical variables were converted into numerical format using one-hot encoding, and irrelevant identifiers such as student IDs were removed. These steps ensured the dataset was clean and suitable for machine learning models.

Methods

The analysis was conducted using Python and followed a structured data mining pipeline. First, the dataset was split into training (80%) and testing (20%) sets to evaluate model performance on unseen data. Feature scaling was applied using standardization to ensure all variables contributed equally to the models.

Two primary techniques were used:

• Linear Regression was applied to predict continuous exam scores. Model performance was evaluated using R^2 Score and Mean Squared Error (MSE). Additionally, 5-fold cross-validation was used to ensure robustness and reduce overfitting.

• Logistic Regression was used to classify students into Pass/Fail categories (threshold = 50%). Model performance was evaluated using an accuracy score and a confusion matrix.

Exploratory Data Analysis (EDA), including correlation heatmaps and scatter plots, was conducted to identify relationships between variables before modelling.

Key Findings & Analysis

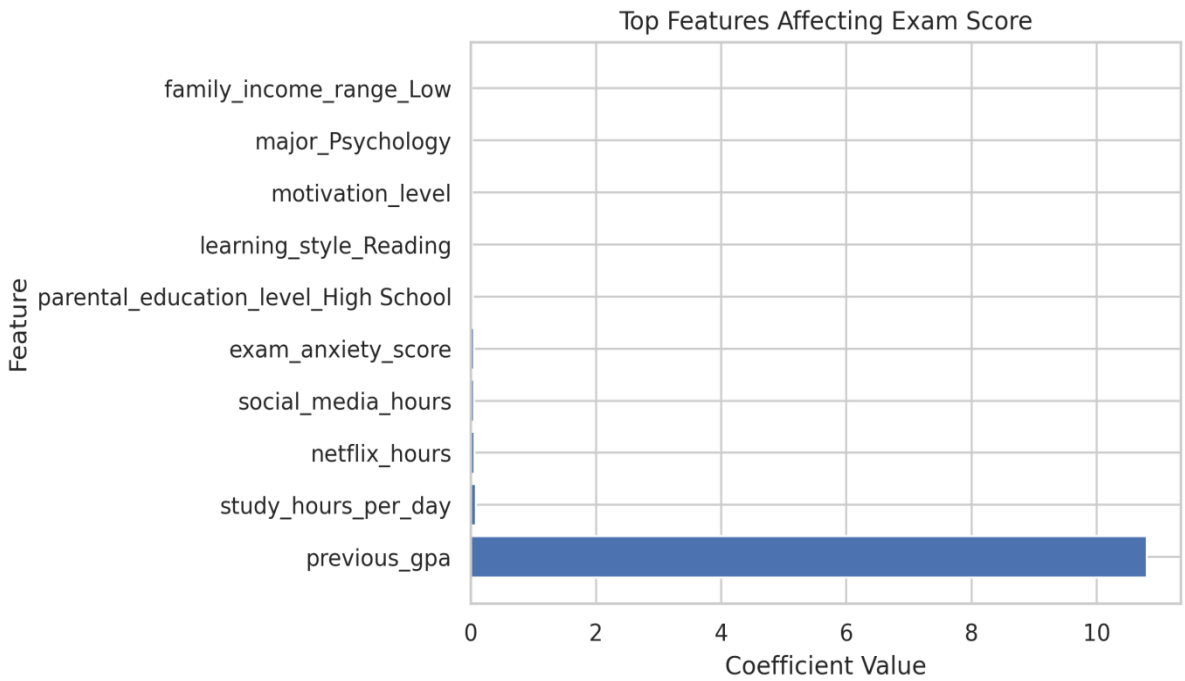
The Linear Regression model showed strong performance, achieving an **R^2 score of 0.8704**, explaining approximately 87% of the variance in exam scores, with a low **MSE of 17.55**. Cross-validation confirmed model robustness, with consistent scores (0.8656–0.8747) and an average R^2 of **0.8702**, indicating good generalization.

A Random Forest model was also applied, achieving a slightly lower **R^2 of 0.8683** and MSE of **17.83**, suggesting that the relationship between variables is largely linear and well captured by Linear Regression.

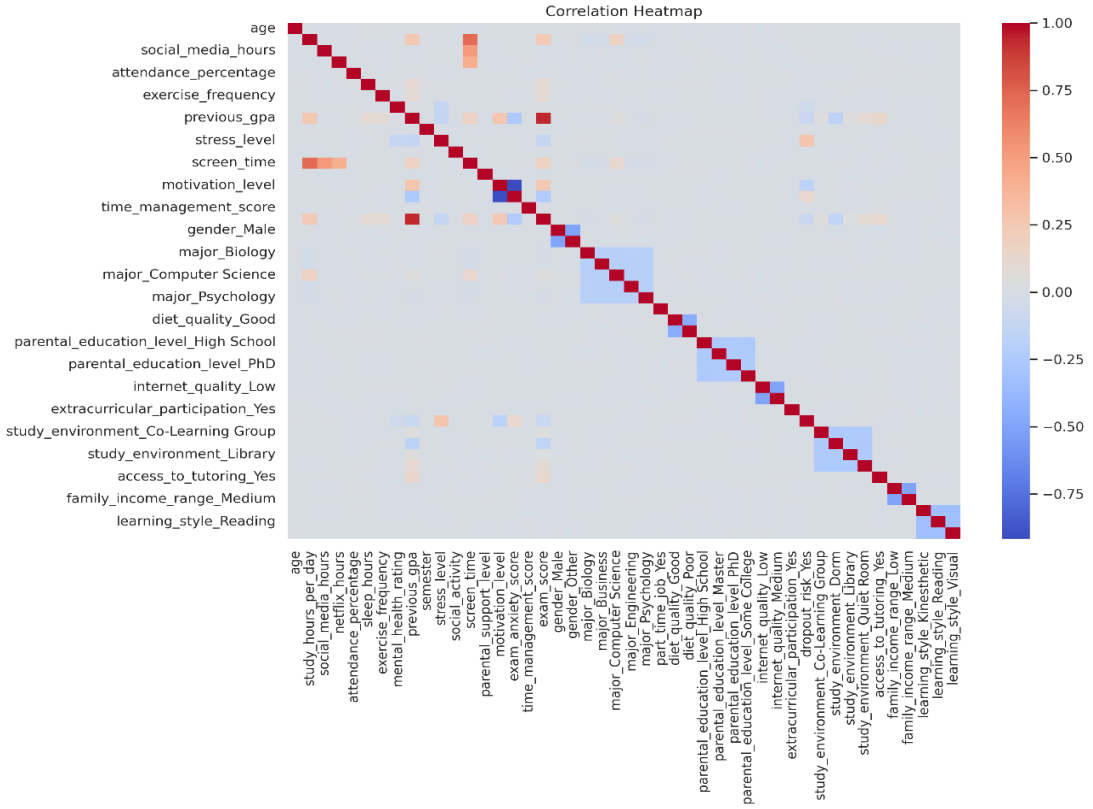
For classification, Logistic Regression achieved a very high **accuracy of 99.83%** in predicting Pass/Fail outcomes. While this indicates strong performance, it may also reflect high class separability or dataset imbalance.

Feature importance analysis identified **study hours, previous GPA, and sleep duration** as key positive predictors, while **stress and screen time** negatively impacted performance. These patterns were supported by exploratory analysis, confirming strong relationships between lifestyle factors and academic outcomes.

Overall, the results demonstrate that student performance can be accurately predicted using behavioural data, with consistent results across multiple models.



Insight: This chart identifies which variables have the strongest impact. Features such as previous_gpa and study_hours_per_day show the highest coefficients, indicating that they are the most reliable predictors of future performance.



Insight: The heatmap highlights critical relationships, such as the negative correlation between stress and scores, allowing the institution to see exactly which lifestyle factors to target for policy changes.

Conclusions

This study shows that student performance can be reliably predicted using lifestyle and behavioural data. Key factors such as study habits, sleep, and stress strongly influence academic outcomes and can act as early indicators of success or failure. Overall, the results highlight that student performance is both predictable and influenced by modifiable behaviours.

Policy Recommendations

Based on these findings, the institution should implement the following data-driven policies:

1. **Early Warning Systems:** Use the classification model to identify "at-risk" students mid-semester based on behavioural indicators like high stress or low attendance.
2. **Mental Health Integration:** Since stress is a primary negative predictor, the institution must increase access to counselling and stress-management workshops.
3. **Holistic Success Workshops:** Shift focus from just "tutoring" to "lifestyle management," educating students on the verified impact of sleep and reduced screen time on their grades.
4. **Targeted Academic Support:** Provide personalized time-management programs for students whose data show high social media usage but low study consistency.